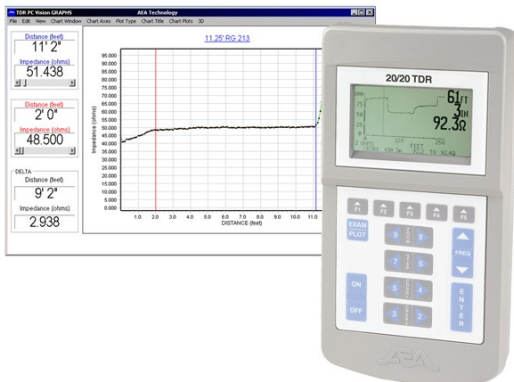


ECE 3317

Dr. Jiefu Chen

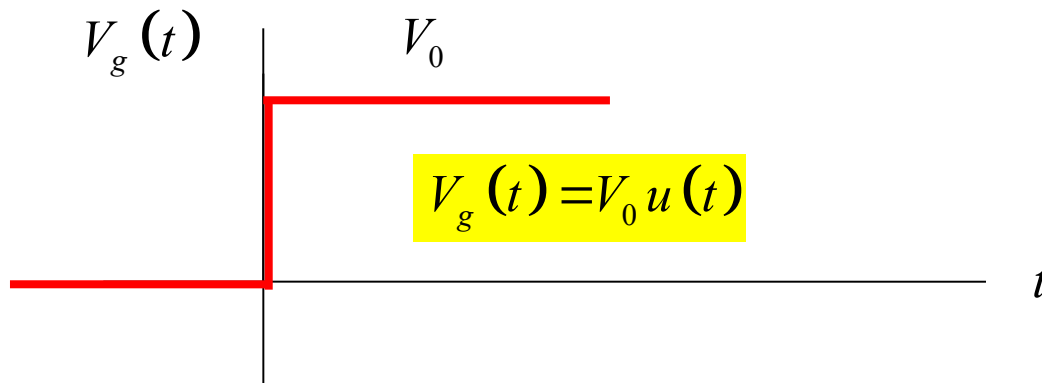
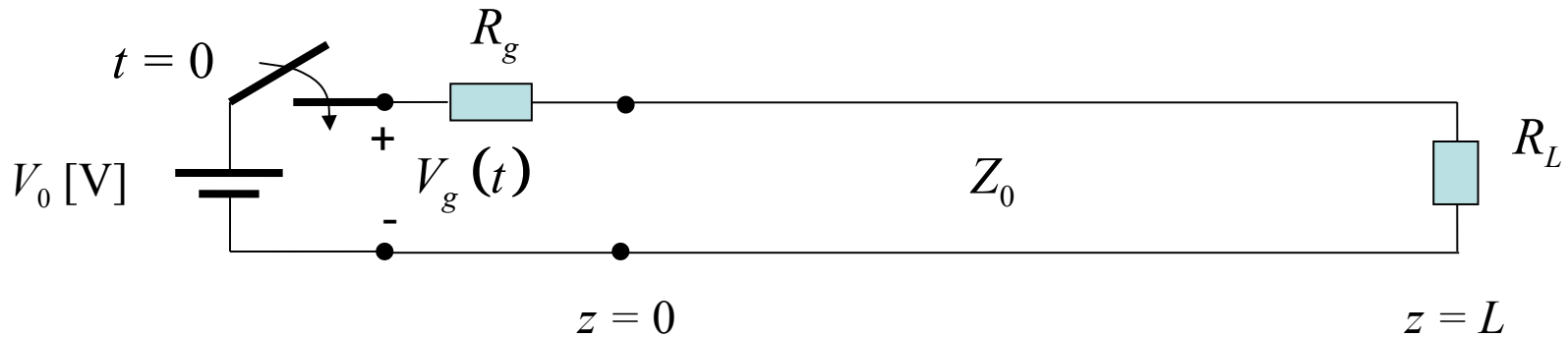
Notes 8 Transmission Lines (Bounce Diagram)



Adapted from notes by Prof. Stuart A. Long

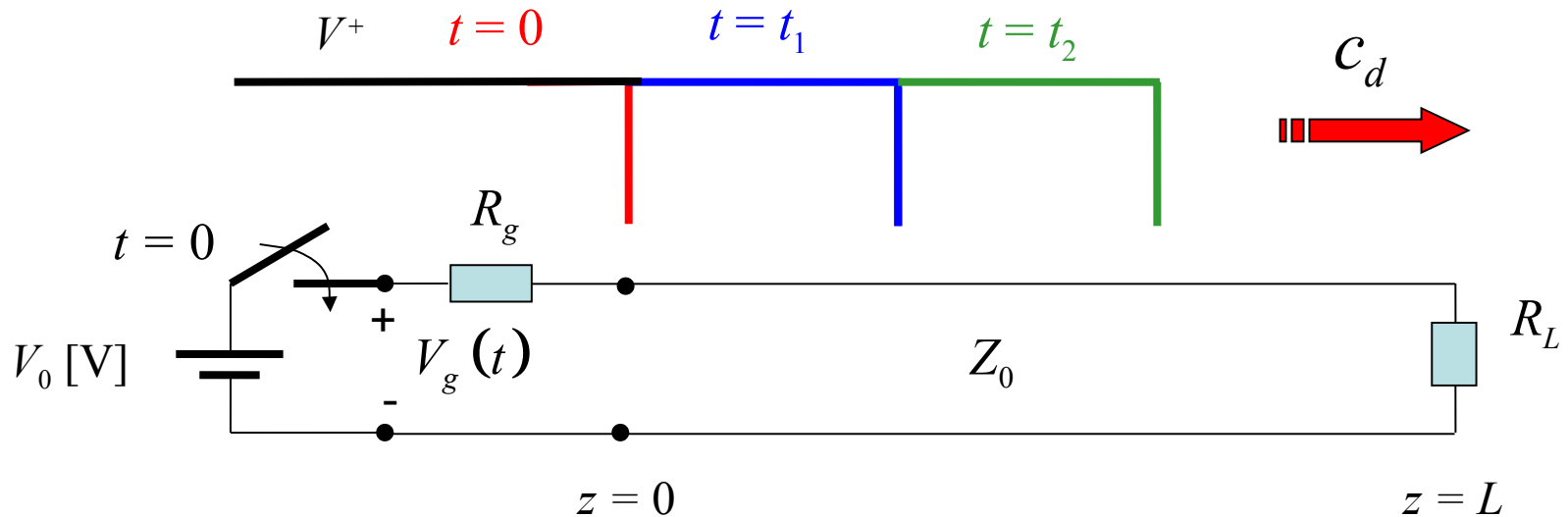
Step Response

The concept of the bounce diagram is illustrated for a unit **step** response on a terminated **lossless** line.



Step Response (cont.)

The wave is shown approaching the load.

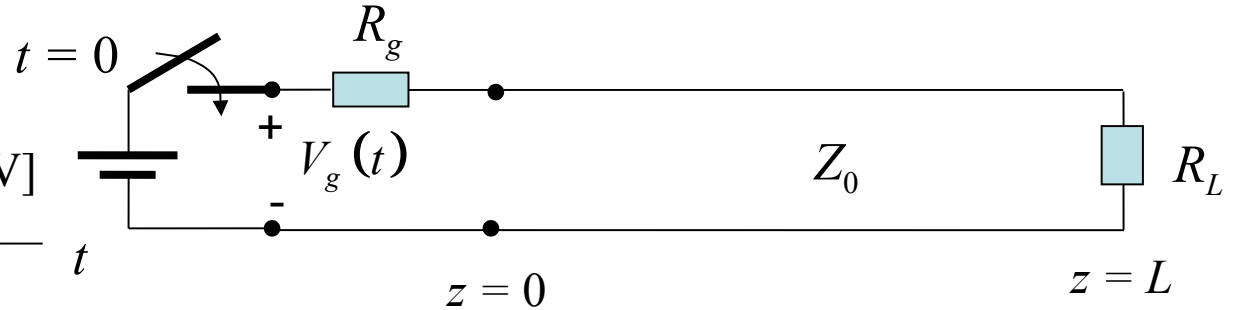
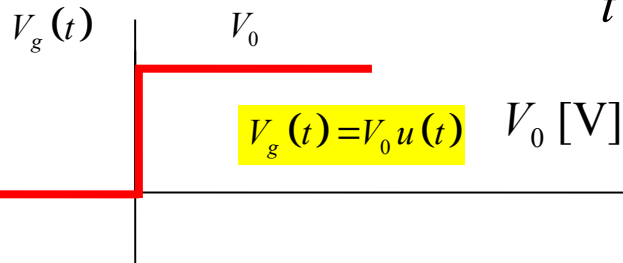


$$V^+ = \left(\frac{Z_0}{R_g + Z_0} \right) V_0$$

(from voltage divider)

Bounce Diagram

Bounce diagram is for a unit step response on a terminated line.

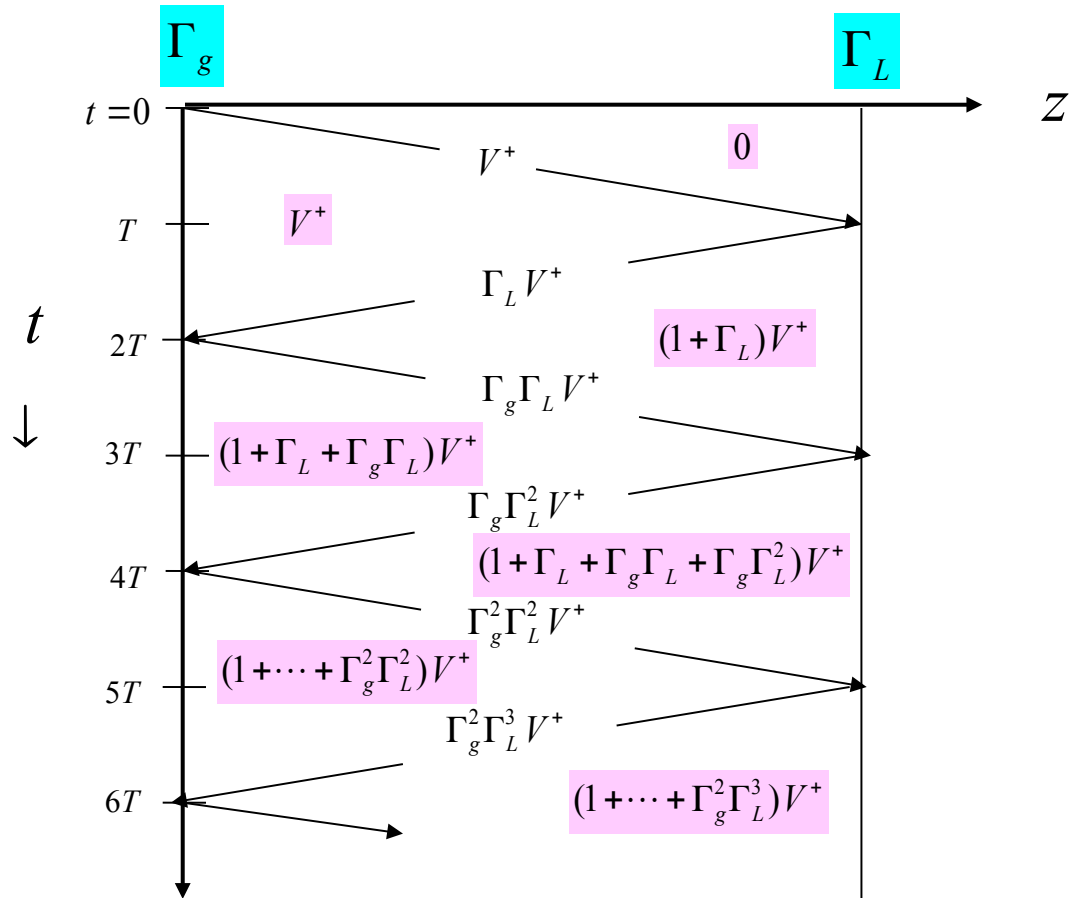


$$T = \frac{L}{c_d}$$

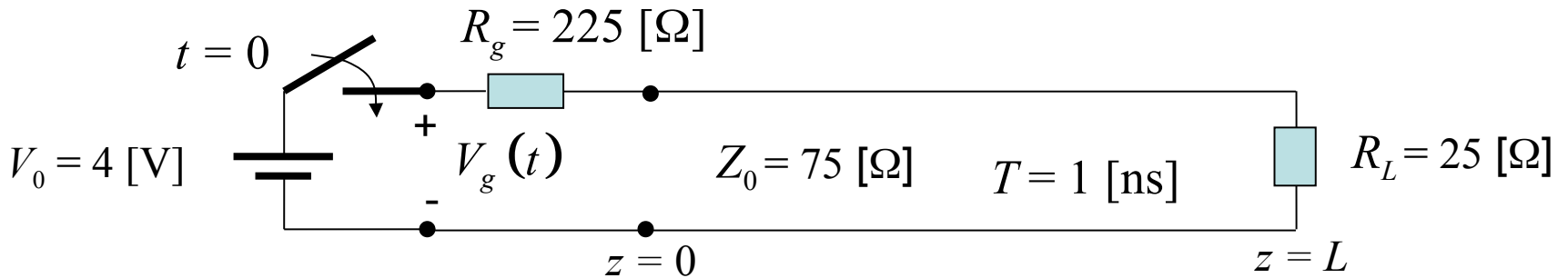
$$V^+ = \left(\frac{Z_0}{R_g + Z_0} \right) V_0$$

$$\Gamma_L = \left(\frac{R_L - Z_0}{R_L + Z_0} \right)$$

$$\Gamma_g = \left(\frac{R_g - Z_0}{R_g + Z_0} \right)$$



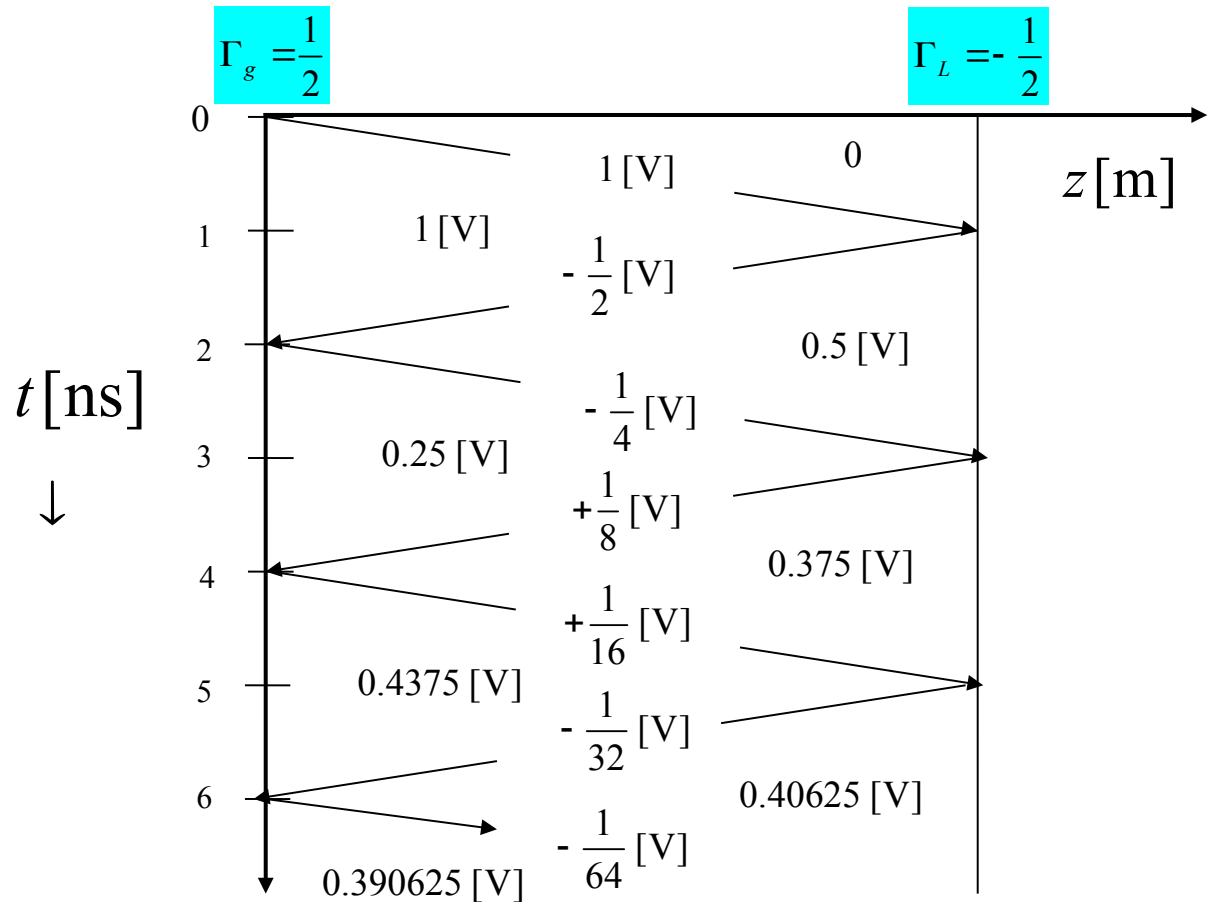
Example



$$V^+ = \frac{Z_0}{R_g + Z_0} V_0 = 1 \text{ [V]}$$

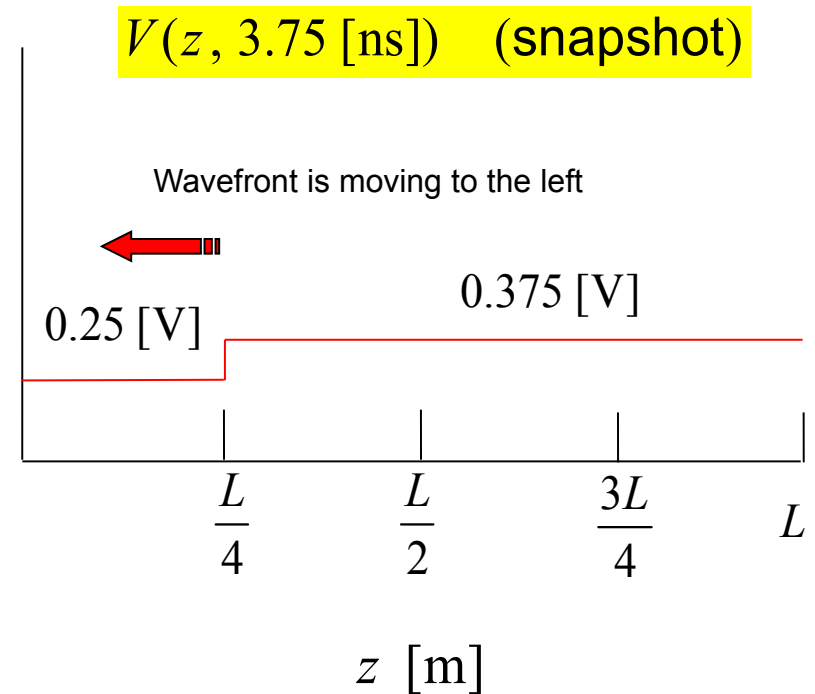
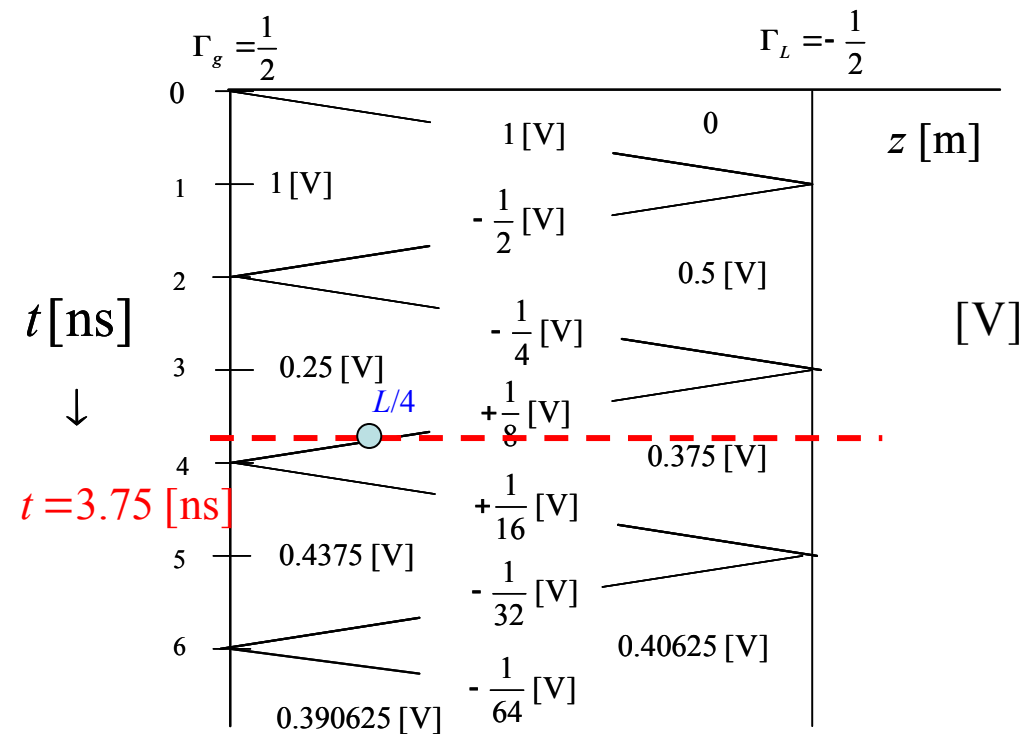
$$\Gamma_L = \left(\frac{R_L - Z_0}{R_L + Z_0} \right) = -\frac{1}{2}$$

$$\Gamma_g = \left(\frac{R_g - Z_0}{R_g + Z_0} \right) = \frac{1}{2}$$



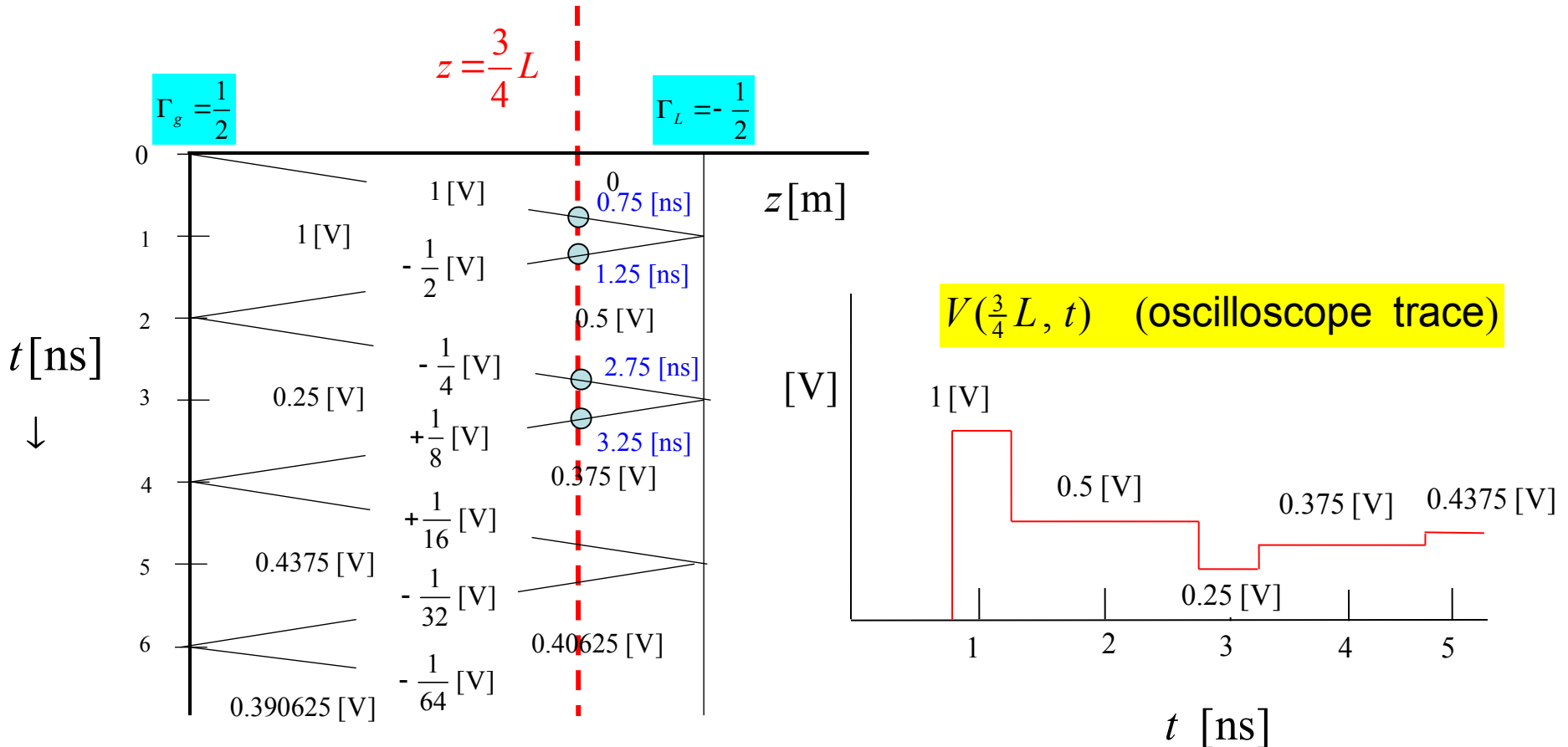
Example (cont.)

The bounce diagram can be used to get a “snapshot” of the line voltage at any point in time.



Example (cont.)

The bounce diagram can be used to get an “oscilloscope trace” at any point on the line.



Steady state voltage: $V(z, \infty) = \left(\frac{R_L}{R_L + R_g} \right) V_0 = 0.400 \text{ [V]}$

Steady-State Solution

Adding all infinite number of bounces, we have:

$$V(z, \infty) = \underbrace{V^+ (1 + \Gamma_g \Gamma_L + \Gamma_g^2 \Gamma_L^2 + \Gamma_g^3 \Gamma_L^3 + \dots)}_{\text{Sum of all right-traveling waves}} + \underbrace{V^+ \Gamma_L (1 + \Gamma_g \Gamma_L + \Gamma_g^2 \Gamma_L^2 + \Gamma_g^3 \Gamma_L^3 + \dots)}_{\text{Sum of all left-traveling waves}}$$

$$= \frac{V^+}{1 - \Gamma_g \Gamma_L} + \frac{V^+ \Gamma_L}{1 - \Gamma_g \Gamma_L} = \frac{(1 + \Gamma_L)}{1 - \Gamma_g \Gamma_L} V^+$$

Note: We have used

$$\sum_{n=0}^{\infty} z^n = \frac{1}{1 - z} \quad (|z| < 1)$$

$$= \frac{\left(1 + \frac{R_L - Z_0}{R_L + Z_0}\right)}{1 - \left(\frac{R_g - Z_0}{R_g + Z_0}\right) \left(\frac{R_L - Z_0}{R_L + Z_0}\right)} \left(\frac{Z_0}{R_g + Z_0} V_0\right)$$

$$= \frac{\left(1 + \frac{R_L - Z_0}{R_L + Z_0}\right) (R_g + Z_0)(R_L + Z_0)}{(R_g + Z_0)(R_L + Z_0) - (R_g - Z_0)(R_L - Z_0)} \left(\frac{Z_0}{R_g + Z_0} V_0\right)$$

Steady-State Solution (cont.)

Simplifying, we have:

$$\begin{aligned}
 V(z, \infty) &= \frac{\left(1 + \frac{R_L - Z_0}{R_L + Z_0}\right) (R_g + Z_0)(R_L + Z_0)}{(R_g + Z_0)(R_L + Z_0) - (R_g - Z_0)(R_L - Z_0)} \left(\frac{Z_0}{R_g + Z_0} V_0 \right) \\
 &= \frac{\left(\frac{2R_L}{R_L + Z_0}\right) (R_g + Z_0)(R_L + Z_0)}{(R_g + Z_0)(R_L + Z_0) - (R_g - Z_0)(R_L - Z_0)} \left(\frac{Z_0}{R_g + Z_0} V_0 \right) \\
 &= \frac{2R_L (R_g + Z_0)}{(R_g + Z_0)(R_L + Z_0) - (R_g - Z_0)(R_L - Z_0)} \left(\frac{Z_0}{R_g + Z_0} V_0 \right) \\
 &= \frac{2R_L Z_0 V_0}{(R_g + Z_0)(R_L + Z_0) - (R_g - Z_0)(R_L - Z_0)} \\
 &= \frac{2R_L Z_0 V_0}{R_g R_L + Z_0^2 + R_L Z_0 + R_g Z_0 - R_g R_L - Z_0^2 + R_L Z_0 + R_g Z_0} \\
 &= \frac{2R_L Z_0 V_0}{+R_L Z_0 + R_g Z_0 + R_L Z_0 + R_g Z_0}
 \end{aligned}$$

Steady-State Solution (cont.)

Continuing with the simplification:

$$\begin{aligned} V(z, \infty) &= \frac{2R_L Z_0 V_0}{R_L Z_0 + R_g Z_0 + R_L Z_0 + R_g Z_0} \\ &= \frac{2R_L Z_0 V_0}{2(R_L Z_0 + R_g Z_0)} \\ &= \frac{R_L Z_0 V_0}{(R_L Z_0 + R_g Z_0)} \\ &= \frac{R_L V_0}{(R_L + R_g)} \end{aligned}$$

Hence we finally have:

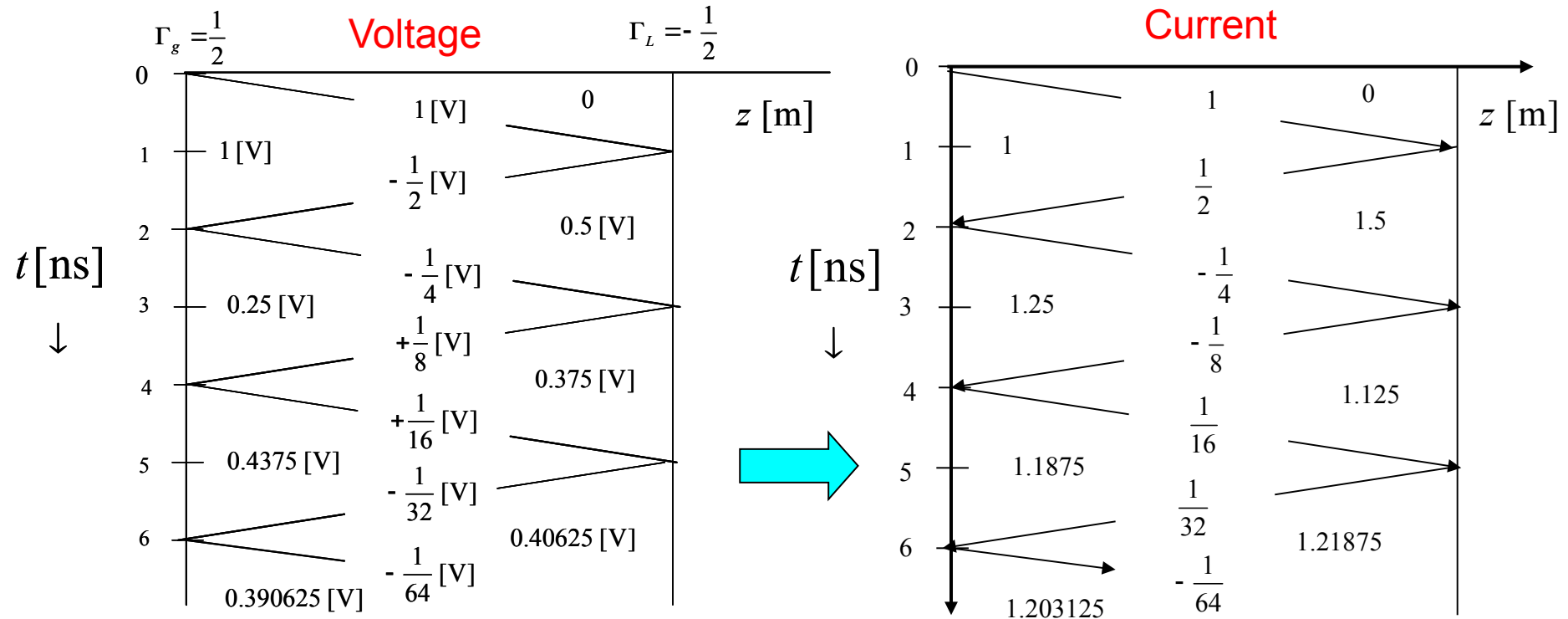
$$V(z, \infty) = \left(\frac{R_L}{R_L + R_g} \right) V_0$$

Note: The steady-state solution does not depend on the transmission line length or characteristic impedance!

This is the DC circuit-theory voltage divider equation!

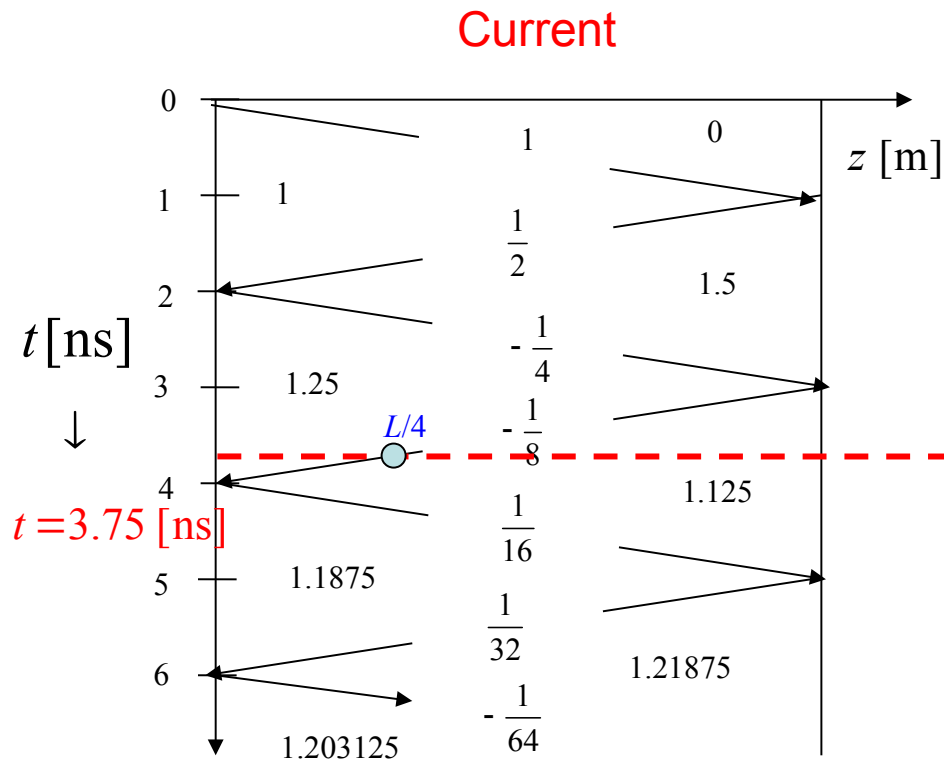
Current Bounce Diagram

To obtain a **current bounce diagram** from the voltage diagram, multiply forward-traveling voltages by $1/Z_0$, backward-traveling voltages by $-1/Z_0$.

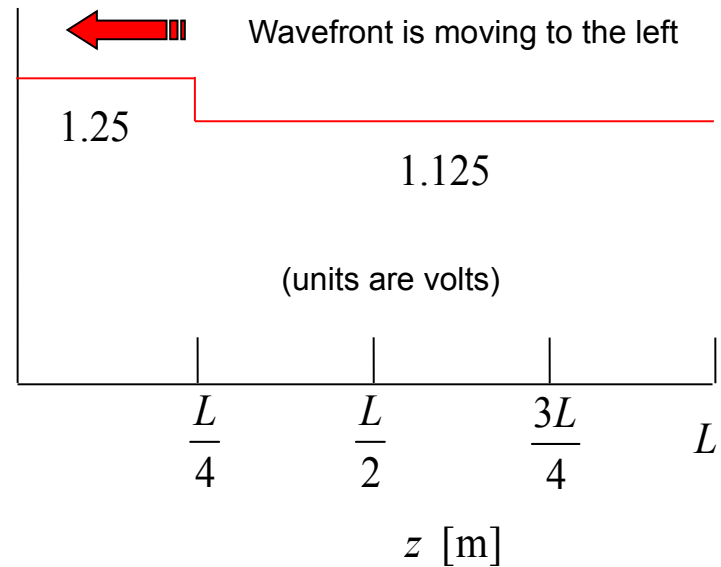


Note: This diagram is for the *normalized* current, defined as $Z_0 I(z, t)$.

Example (cont.)

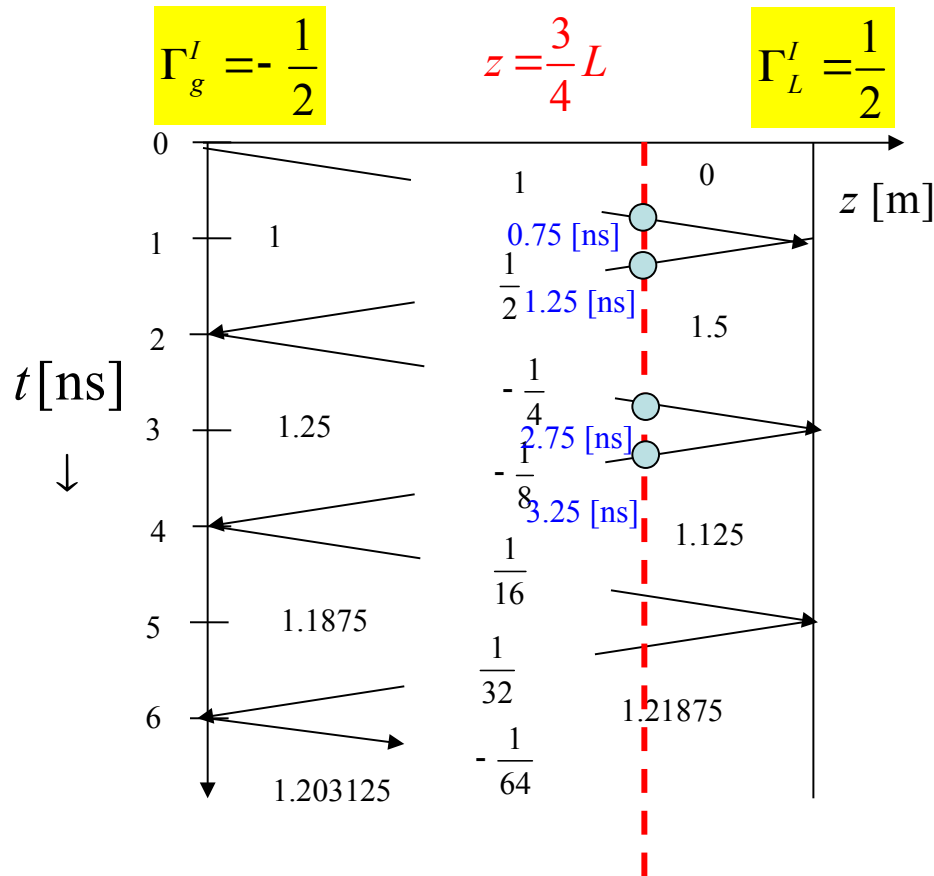


$Z_0 I(z, 3.75 \text{ [ns]})$
(snapshot of current)

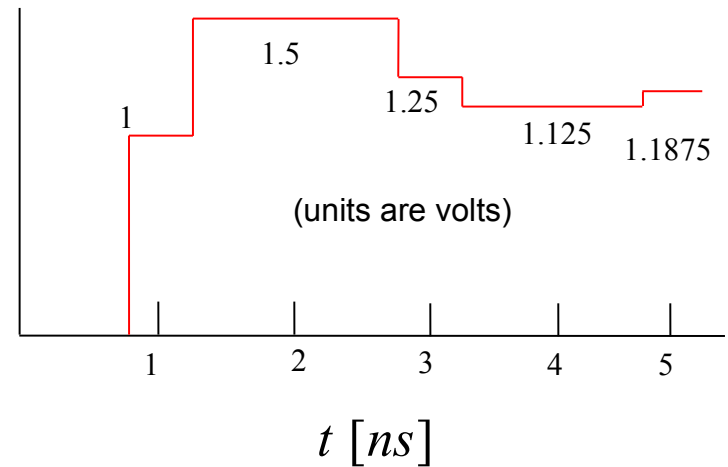


Example (cont.)

Current



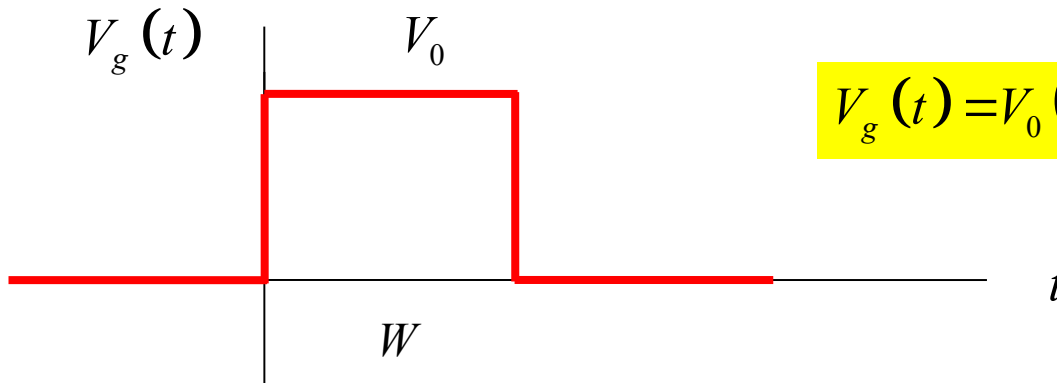
$Z_0 I(\frac{3}{4}L, t)$
(oscilloscope trace of current)



Steady state current: $I(z, \infty) = \left(\frac{V_0}{R_L + R_g} \right) = 0.016 \text{ [A]}$ $Z_0 I(z, \infty) = (0.016)(75) = 1.20$

Pulse Response

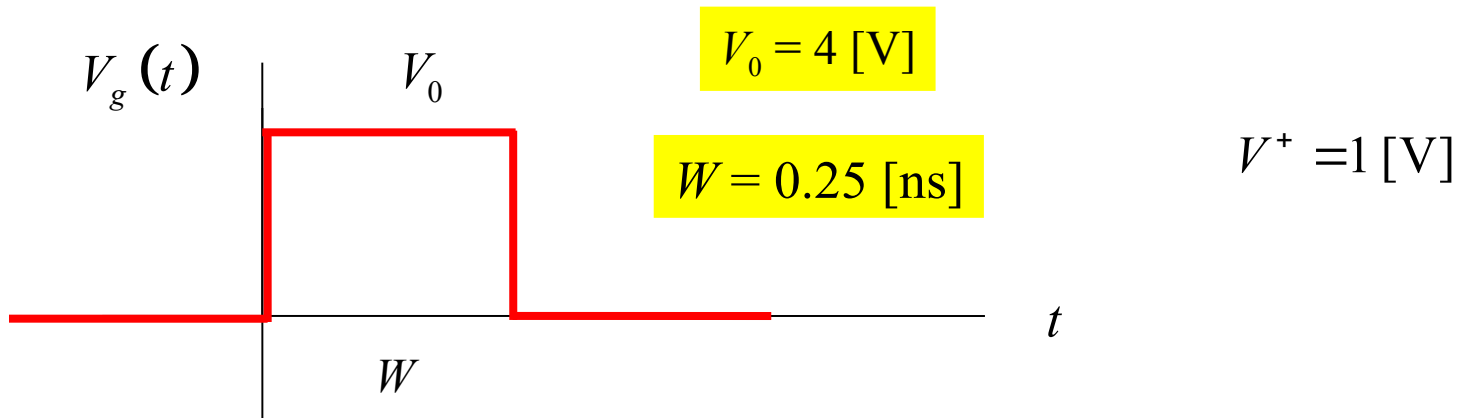
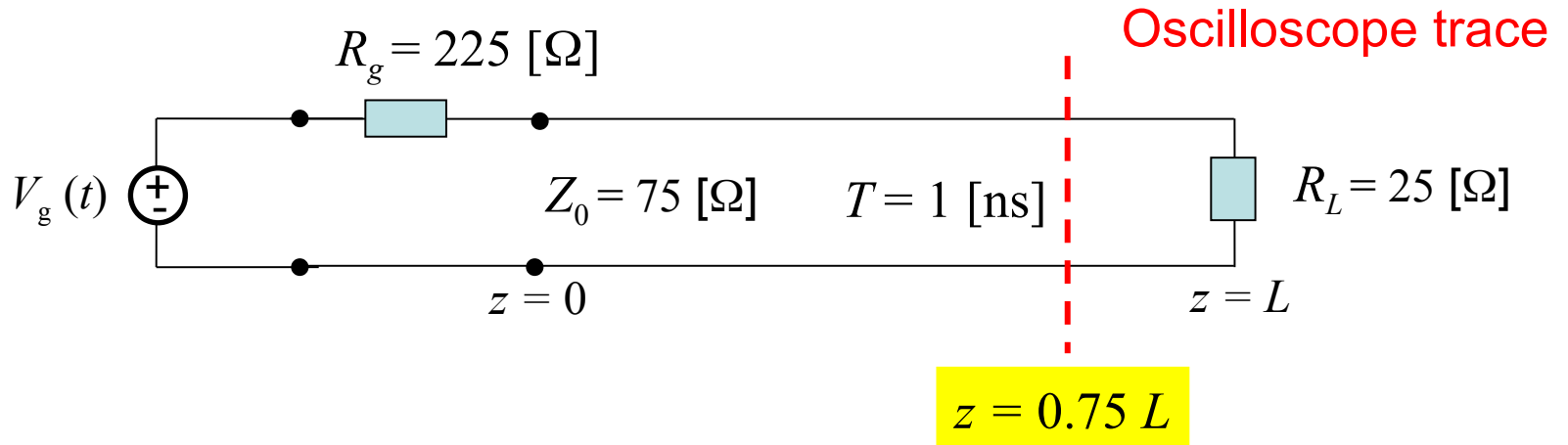
Superposition can be used to get the response due to a pulse.



$$V_g(t) = V_0(u(t) - u(t - W))$$

We thus subtract two bounce diagrams, with the second one being a shifted version of the first one.

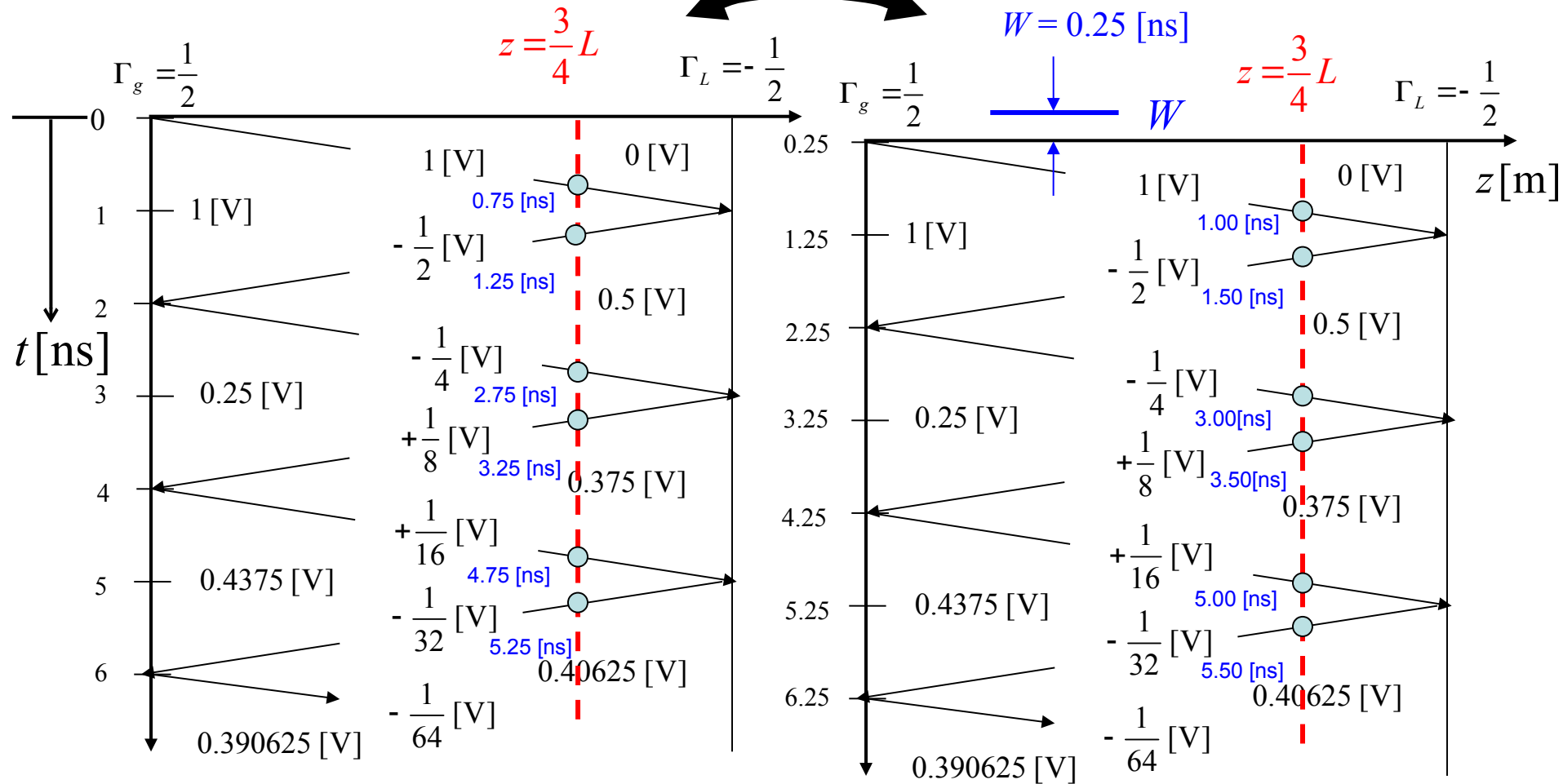
Example: Pulse



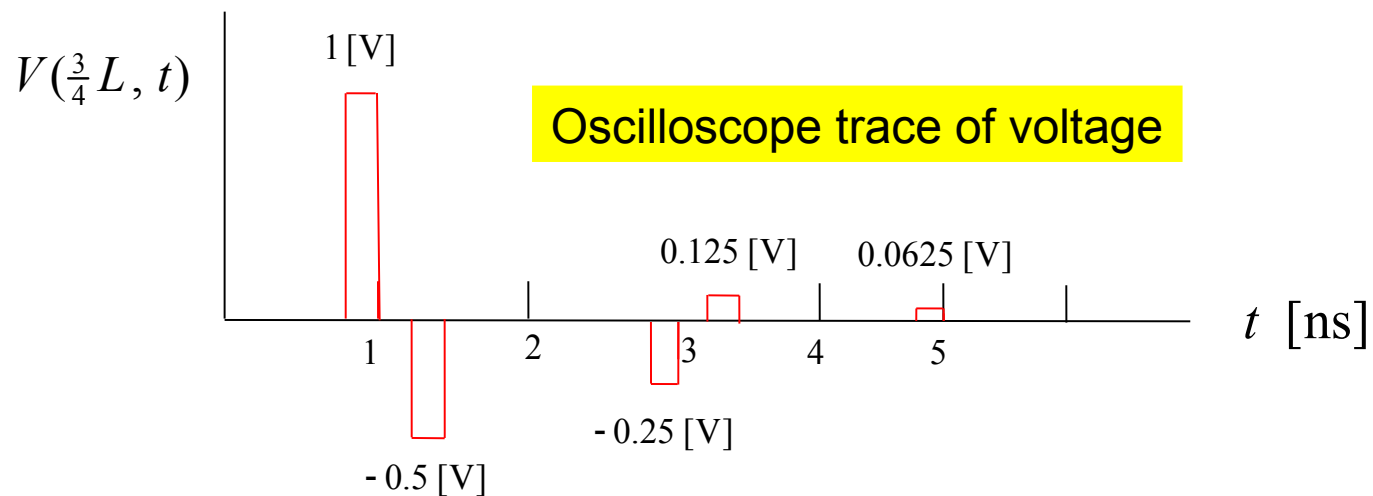
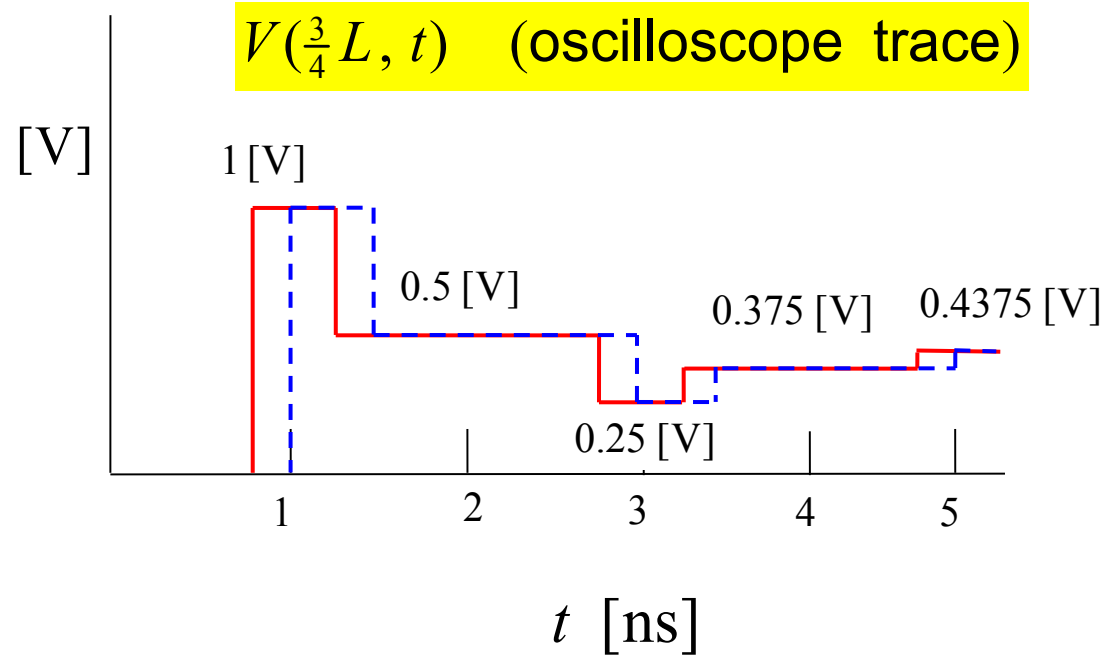
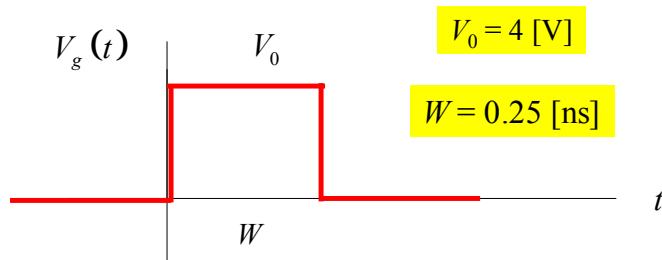
Example: Pulse (cont.)

Subtract

$$V_g(t) = V_0(u(t) - u(t - W))$$



Example: Pulse (cont.)



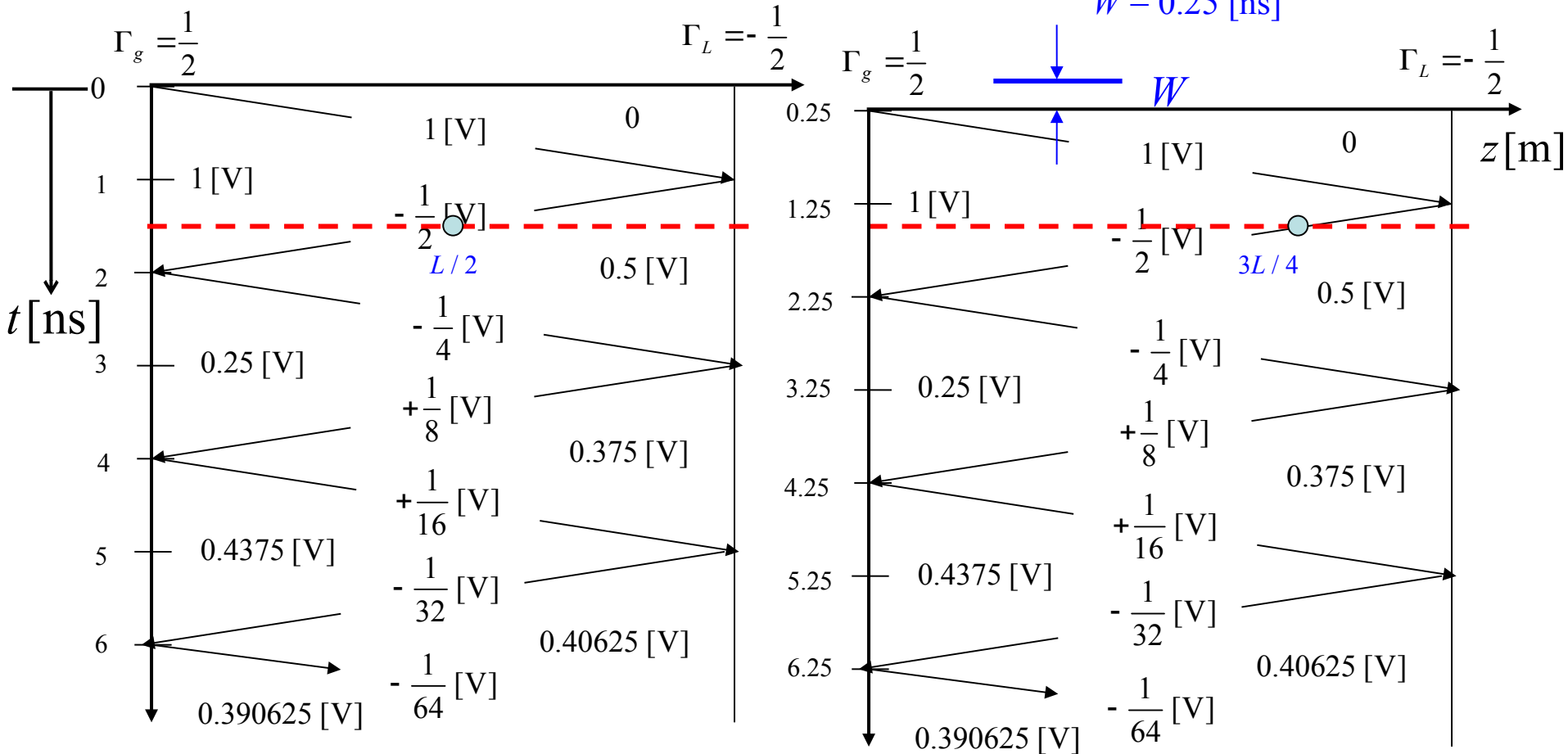
Example: Pulse (cont.)

$$V_g(t) = V_0(u(t) - u(t - W))$$

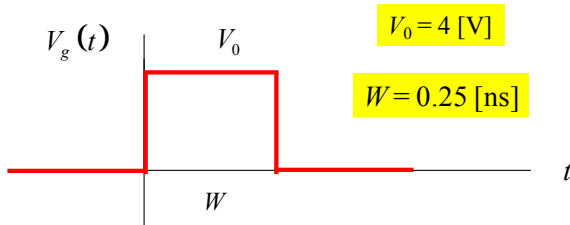
subtract

Snapshot

$$t = 1.5 \text{ [ns]}$$

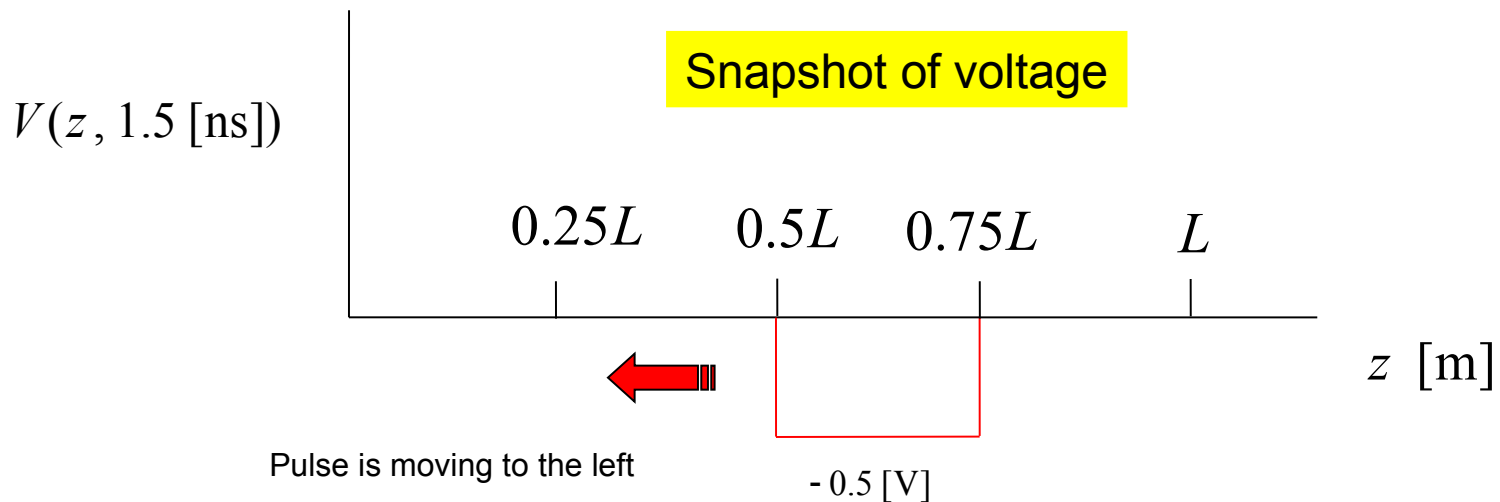
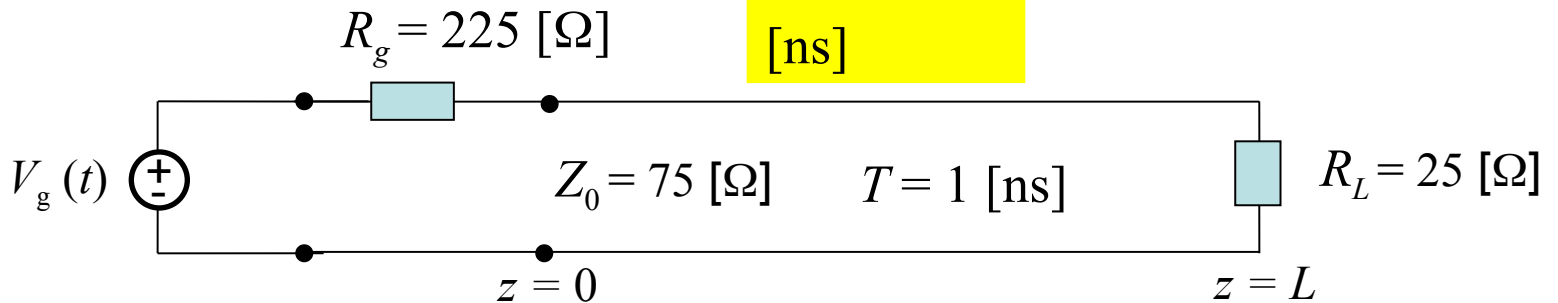


Example: Pulse (cont.)



$$V_0 = 4 \text{ [V]}$$

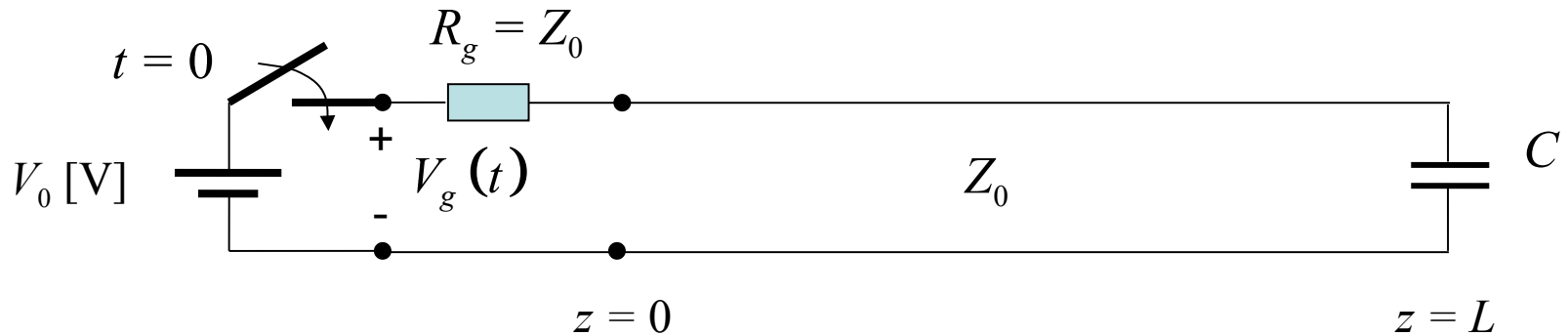
$$W = 0.25 \text{ [ns]}$$



Pulse is moving to the left

$$-0.5 \text{ [V]}$$

Capacitive Load



Note: The generator is assumed to be matched to the transmission line for convenience (we wish to focus on the effects of the capacitive load).

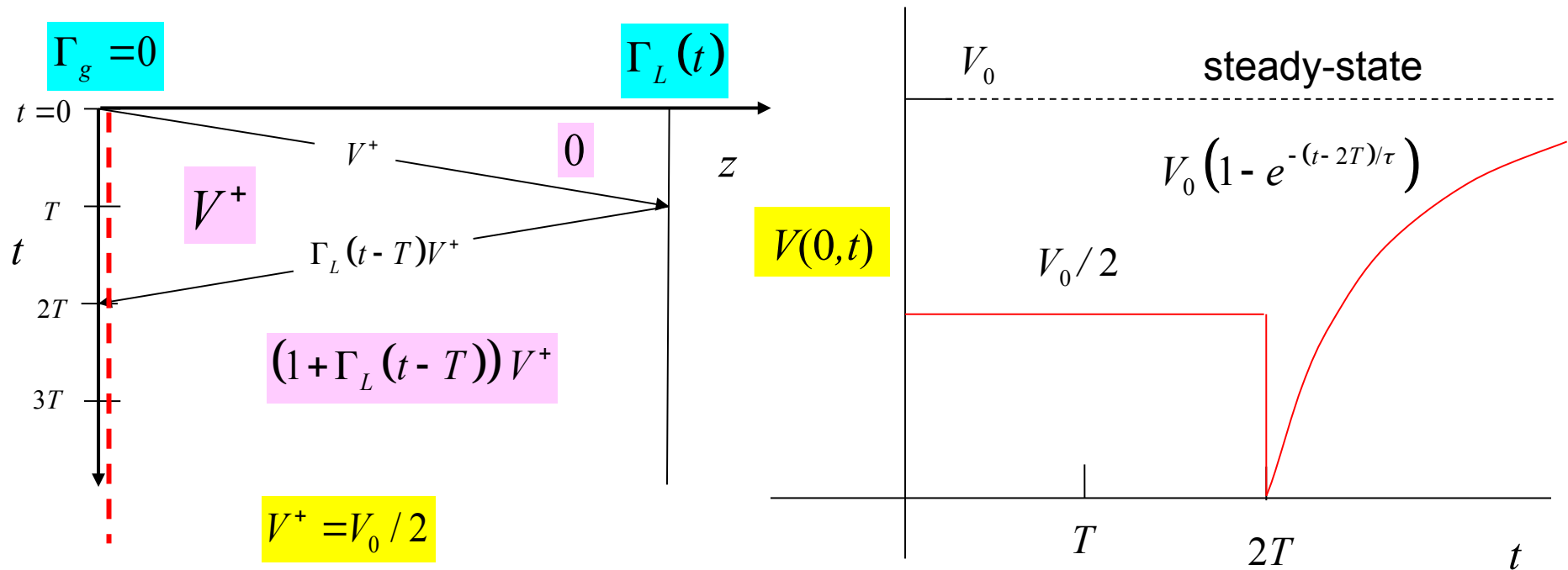
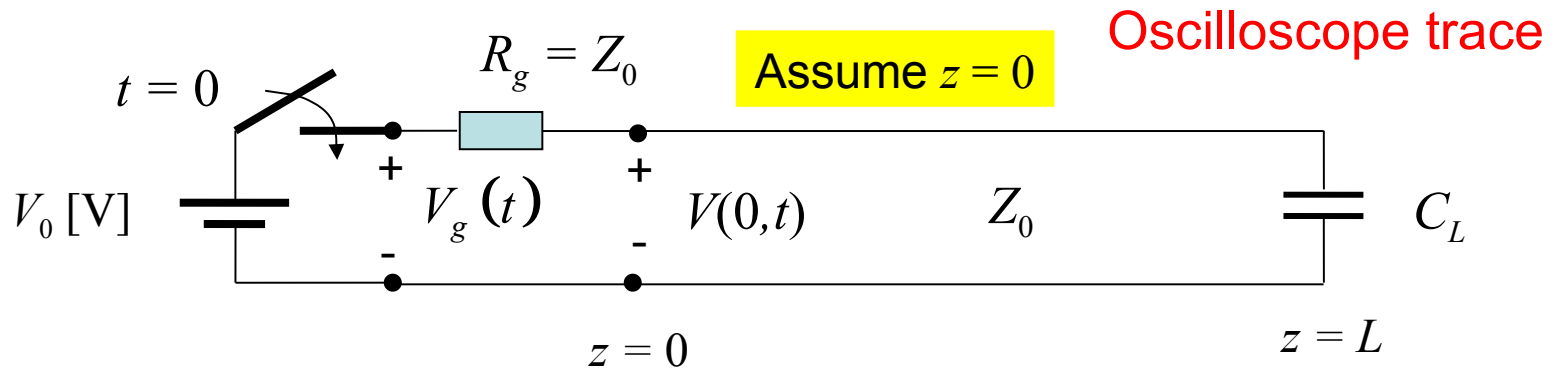
$$\text{Hence } \Gamma_g = 0, \quad V^+ = \left(\frac{Z_0}{Z_0 + Z_0} \right) V_0 = V_0 / 2$$

At $t = T$: The capacitor acts as a **short circuit**: $\Gamma_L(T) = -1$, $V(L, T) = 0$

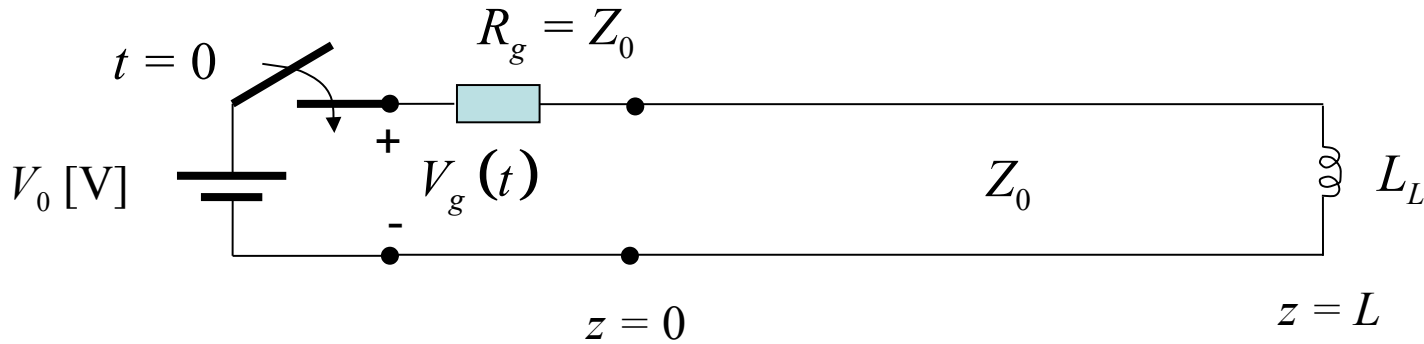
At $t = \infty$: The capacitor acts as an **open circuit**: $\Gamma_L(\infty) = 1$, $V(L, \infty) = V_0$

Between $t = T$ and $t = \infty$, there is an exponential time-constant behavior.

Capacitive Load (cont.)



Inductive Load

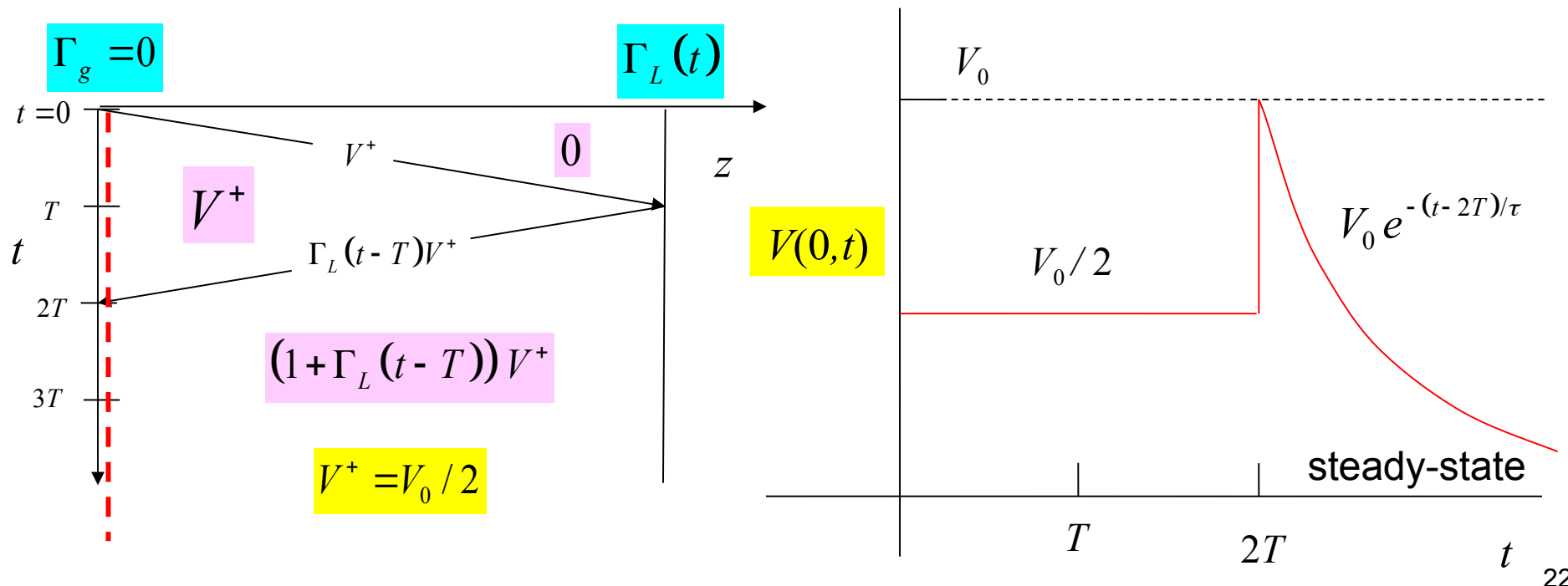


At $t = T$: inductor as a **open circuit**:

$$\Gamma_L(T) = 1, V(L, T) = 0$$

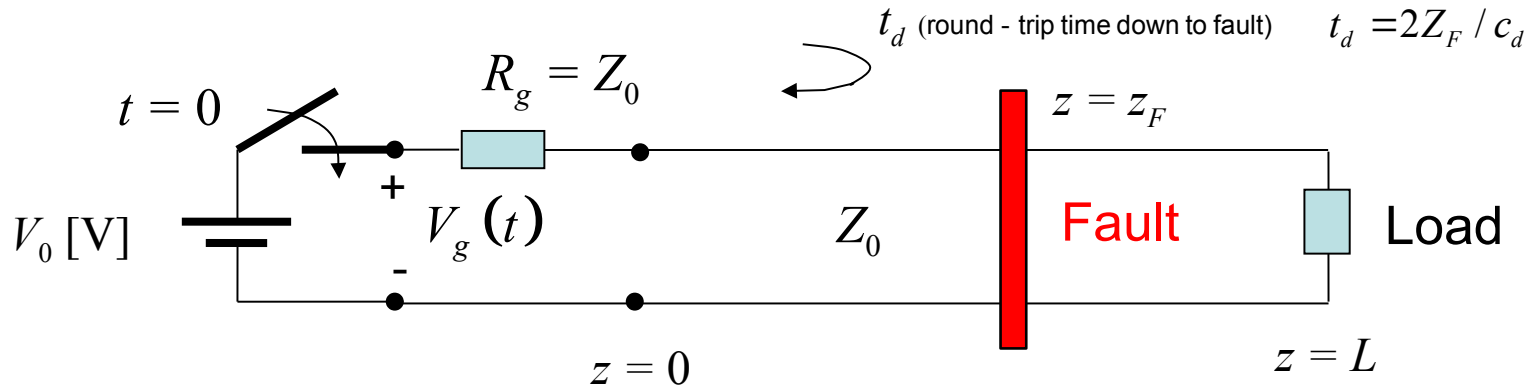
At $t = \infty$: inductor acts as a **short circuit**:

$$\Gamma_L(\infty) = -1, V(L, \infty) = 0$$

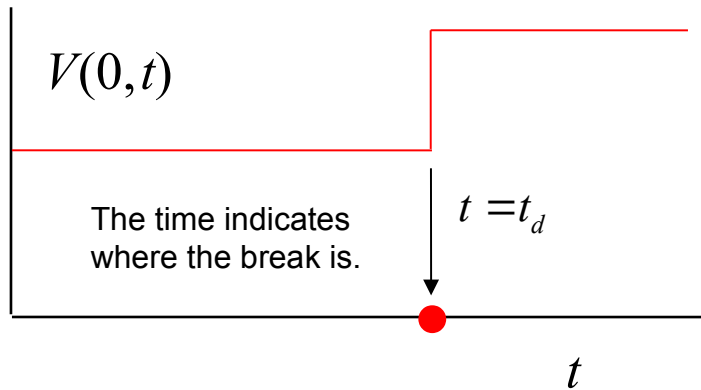


Time-Domain Reflectometer (TDR)

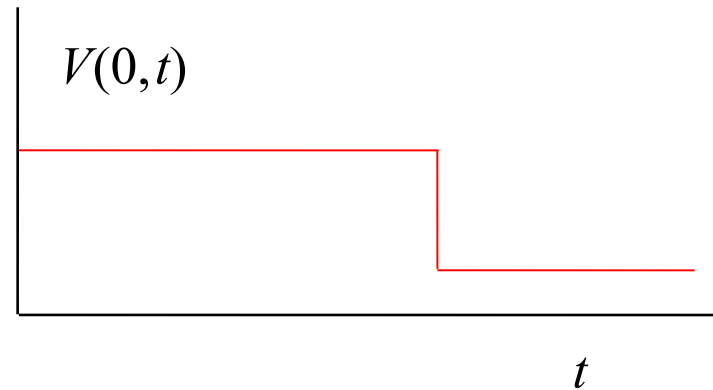
This is a device that is used to look at reflections on a line, to look for potential problems such as breaks on the line.



The fault is modeled as a load resistor at $z = z_F$.



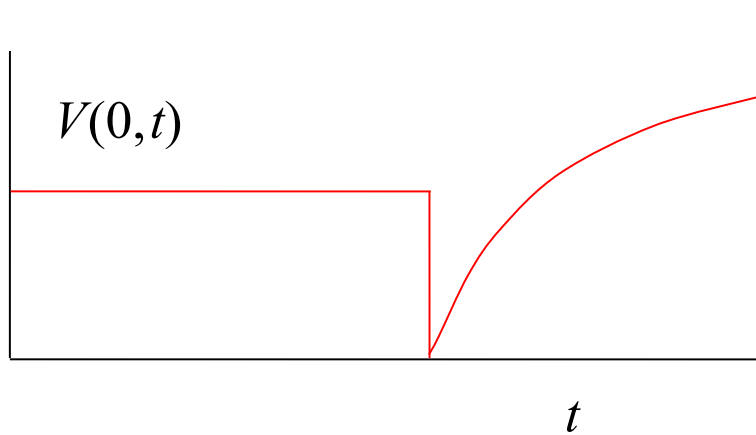
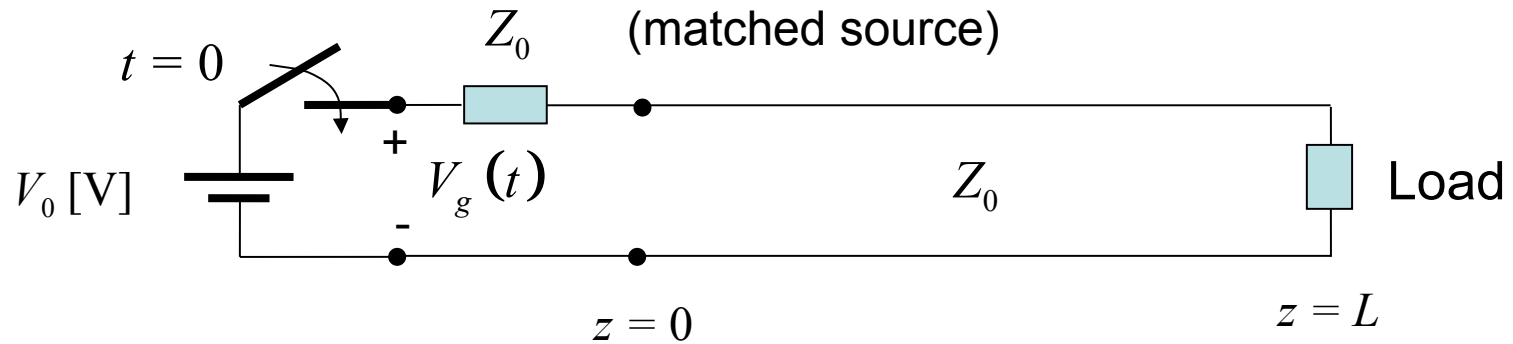
resistive load, $R_L > Z_0$



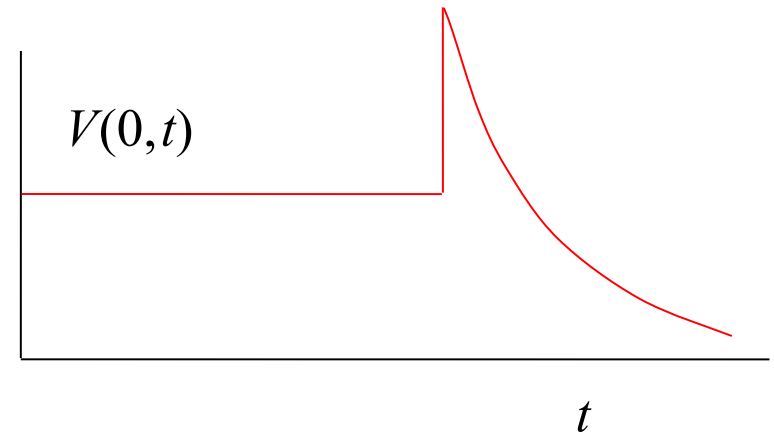
resistive load, $R_L < Z_0$

Time-Domain Reflectometer (cont.)

The reflectometer can also tell us what kind of a load we have.



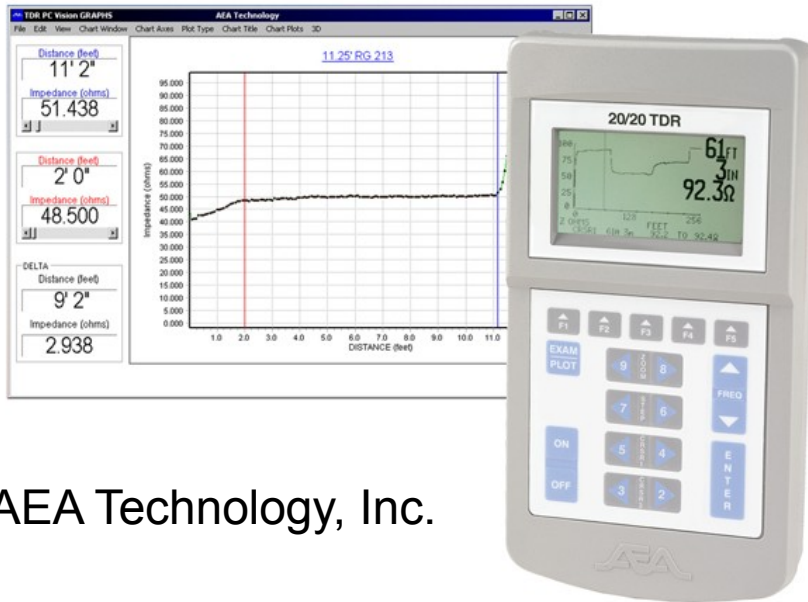
Capacitive load



Inductive load

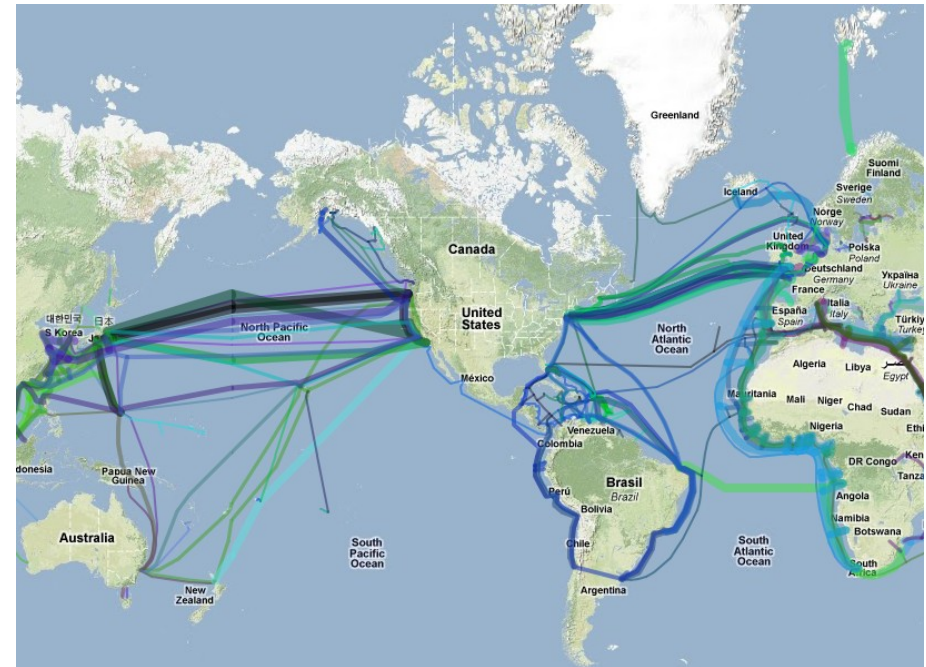
Time-Domain Reflectometer (cont.)

ritholtz.com



AEA Technology, Inc.

example of a commercial product



submarine communication cable